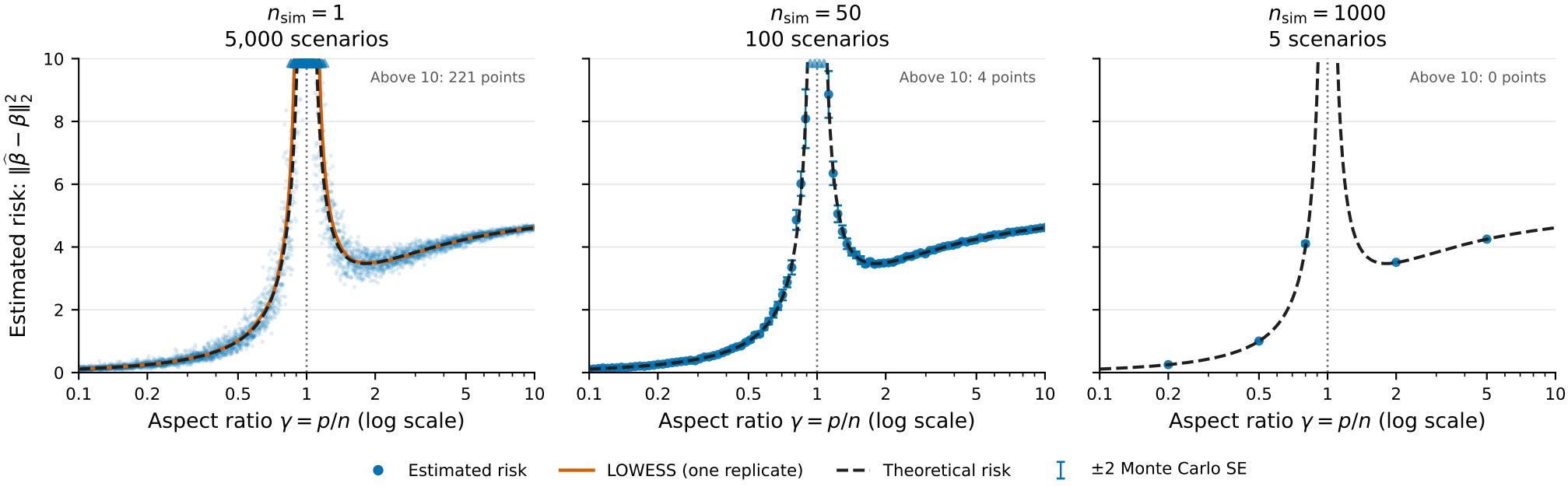

Ridgeless least squares: $n = 200$, $r^2 = 5$
$n_{\text{sim}} = 1$
5,000 scenarios
$n_{\text{sim}} = 50$
100 scenarios
$n_{\text{sim}} = 1000$
5 scenarios
Above 10: 221 points
Above 10: 4 points
Above 10: 0 points
Estimated risk: $\|\widehat{\beta} - \beta\|_2^2$
Aspect ratio $\gamma = p/n$ (log scale)
Estimated risk
LOWESS (one replicate)
Theoretical risk
±2 Monte Carlo SE
LOWESS: frac=0.1, it=0; separate fits for gamma < 1 and gamma > 1.
View limited to risk <= 10; triangles mark estimates above this limit. Raw values are retained.